23. Wednesday, May 21. Bijections (6.3) Composition. (6.4)

Last Time: Injections (aka one-to-one)
Surjections (a.ka onto)

Examples: Recall $\mathbb{Z}_5 = \{0,1,2,3,4\}$ with addition and mult. modulo 5.

1) $f: \mathbb{Z}_5 \longrightarrow \mathbb{Z}_5 \quad f(x) = x^2$

Note $f(0)=0$, $f(1)=1$, $f(2)=4$, $f(3)=4$, $f(4)=1$
Not injective, because $f(1) = f(4)$
Not surjective, because $\nexists\, x \in \mathbb{Z}_5$ such that $f(x)=2$.

2) $g: \mathbb{Z}_5 \longrightarrow \mathbb{Z}_5$, $g(x) = x^3$

Note $g(0)=0$, $g(1)=1$, $g(2)=3$, $g(3)=2$, $g(4)=4$

Injective, because $g(x) \neq g(y)$ whenever $x \neq y$
Surjective. because g hits all elements in $\mathbb{Z}_5$.

Theorem 1: The function $h: \mathbb{R}^2 \to \mathbb{R}^2$ $h(x,y) = (x+y, x-y)$ is bijective.

Proof. We prove h is injective and surjective separately

[Injectivity] We need to show that, for any $a, b \in \mathbb{R}^2$, if $h(a) = h(b)$, then $a = b$.

Suppose $h(x_1, y_1) = h(x_2, y_2)$ for some (x_1, y_1) and $(x_2, y_2) \in \mathbb{R}^2$.

This means $(x_1+y_1, x_1-y_1) = (x_2+y_2, x_2-y_2)$. Then $x_1+y_1 = x_2+y_2$ and $x_1-y_1 = x_2-y_2$

Adding these last two, we get $2x_1 = 2x_2$, so $x_1 = x_2$.

Subtracting first - second yields $2y_1 = 2y_2$, so $y_1 = y_2$

Hence $(x_1, y_1) = (x_2, y_2)$, as desired.

[Surjectivity] We need to show that for every $b \in \mathbb{R}^2$ exists $a \in \mathbb{R}^2$ such that $h(a) = b$

Let $(c,d) \in \mathbb{R}^2$. We need to find some $(x,y) \in \mathbb{R}^2$ such that $h(x,y) = (c,d)$

[Scratch work want: $(x+y, x-y) = (c,d)$, so $x+y = c \rightsquigarrow 2x = c+d \rightsquigarrow x = \frac{c+d}{2}$

$x-y = d \rightsquigarrow 2y = c-d \rightsquigarrow y = \frac{c-d}{2}$

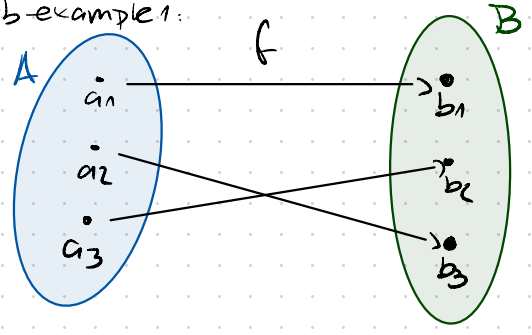
Let $(x,y) = \left(\frac{c+d}{2}, \frac{c-d}{2}\right)$ Then

$h(x,y) = (x+y, x-y) = \left(\frac{c+d}{2} + \frac{c-d}{2}, \frac{c+d}{2} - \frac{c-d}{2}\right) = (c,d)$. Since (c,d) is arbitrary, h is surj. □

Example: Let $f: A \to B$ be any function. Define $g: B \to \mathcal{P}(A)$ by $g(y) = f^{-1}(\{y\})$ → Assume A is finite

Is g injective? Is it surjective?

Sub-example 1:



$g: B \to \mathcal{P}(A)$

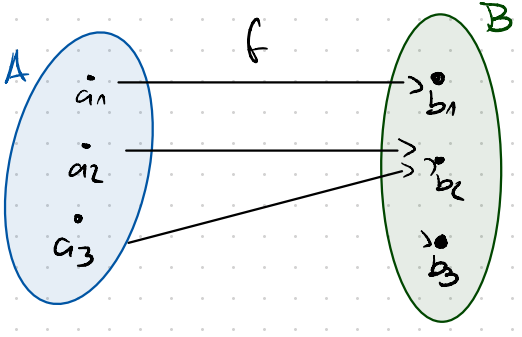
$g(b_1) = \{a_1\}$

$g(b_2) = \{a_3\}$

$g(b_3) = \{a_2\}$

For this particular f,
g injective
g not surjective

Sub-example 2



$g(b_1) = \{a_1\}$

$g(b_2) = \{a_2, a_3\}$

$g(b_3) = \phi$

Again,
g injective
g not surjective

It seems like in general is hard for g to be surjective and hard for g to not be injective.

General fact: Since f is a function, for each $a \in A$ there is a unique $b \in B$ such that $a \in g(b)$.

In other words, $A = \bigcup_{X \in g(B)} X$ and the union is disjoint.

So for g to be surjective we would need $A = \bigcup_{X \in \mathcal{P}(A)} X$ with a disjoint union!!

On the other hand, for injectivity, if $g(b_1) = g(b_2)$ then these sets must be empty!

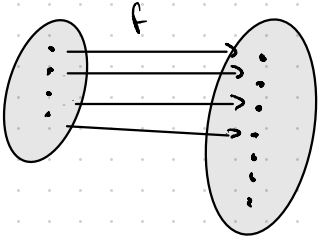
Exercise: Prove the following

1) g is surjective if and only if $|A| = 1$ and $|B| \geq 2$.

2) g is injective if and only if ($f(A) = B$ or $f(A) = B \setminus \{\text{a single element}\}$).

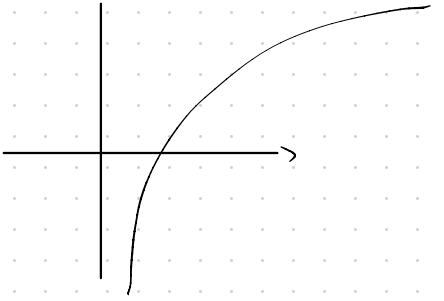
Question Suppose $A \nsim B$. Is it possible to find a bijection $f: A \longrightarrow B$?

Answer If A and B are finite, the answer is no,

any function 

Will miss some elements of B

Now, if A and B are infinite, then it's possible.

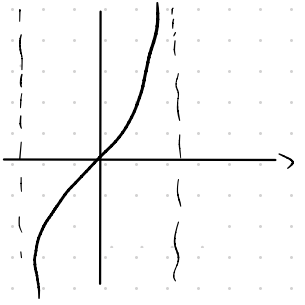


Examples: 1) Let $A = \mathbb{R}^+ = [0, \infty)$ and $B = \mathbb{R}$.

Then $f: A \longrightarrow B$, $f(x) = \ln(x)$ is a bijection

2) Further, $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \longrightarrow \mathbb{R}$

$f(x) = \tan(x)$ is a bijection



3) $A = \{\text{Even integers}\}$, $B = \mathbb{Z}$

Then $\ell: A \to B$, $\ell(x) = \frac{x}{2}$ is a bijection

4) $A = \mathbb{N}$, $B = \mathbb{Z}$. Find a bijection $f: A \to B$.

5) $A = \mathbb{N}$, $B = \mathbb{R}$. $\exists$ bijection $f: A \to B$?

Composition

Definition 2: Let A, B, C be nonempty sets and let $f: A \to B$ and $g: B \to C$.

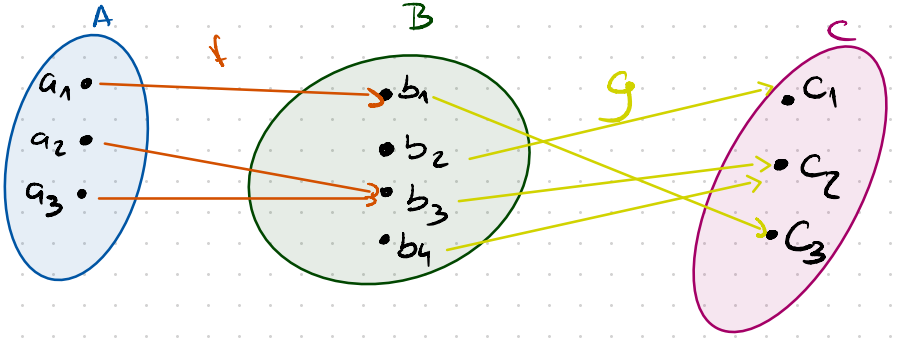
The **composition** of f and g is the function $g \circ f: A \to C$ defined by $(g \circ f)(a) = g(f(a))$ for any $a \in A$.

↳ "circ"

read "g of f"

We refer to $g \circ f$ as a composite function

Example:



$(g \circ f)(a_1) = g(f(a_1)) = g(b_1) = c_3$,

$(g \circ f)(a_2) = c_2$, $(g \circ f)(a_3) = c_2$.

Theorem 3: Let $f: A \to B$ and $g: B \to C$. The following hold:

1) If $g \circ f$ is injective then f is injective.

2) If $g \circ f$ is surjective then g is surjective.

Proof:

1) Suppose $g \circ f$ is injective. We want to prove that f is injective.

Thus, let $x, y \in A$ such that $f(x) = f(y)$. We want to show $x = y$.

Since $f(x) = f(y)$, we can apply g to get $g(f(x)) = g(f(y))$. By def^n of composition, this means $(g \circ f)(x) = (g \circ f)(y)$. Since $g \circ f$ is injective and $(g \circ f)(x) = (g \circ f)(y)$, it must be $x = y$, as desired.

2) Exercise

□

Exercise: Give another proof for 1 by contrapositive.

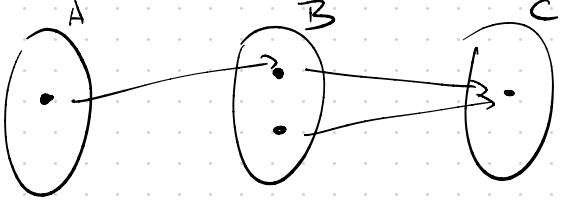
Exploration: Assume $g \circ f$ is surjective.

This says: for every $c \in C$, there is some $a \in A$ such that

$$\underbrace{(g \circ f)(a)}_{g(f(a))} = c$$

So, $\forall c \in C$, $\exists\, b \in B$ (namely $b = f(a)$) such that $g(b) = c$.

Can we say more? Let's do some examples

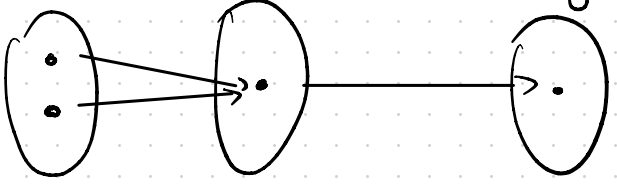


Here $g \circ f$ is surjective,

f is injective, not surjective

and g is not injective

Does f have to be injective?



$g \circ f$ surjective

g injective

f surjective, f not injective

In conclusion. If $g \circ f$ is surjective, then.

- g is surj.
- g may or may not be inj.
- f " surj.
- f " inj.

Exercise: Assume $g \circ f$ is injective. Do a similar exploration.